

Skills Summary

Quotients and Remainders with Models

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading a Quotient off a Model

Every division fits one sentence:

total = (number of groups) $\times$ (size of a group) + remainder.

On a model, the **quotient** is what you counted — full groups, or how many are in each group — and the **remainder** is what would not fit into any group.

The remainder is always *smaller* than the group size. If it is not, one more group can still be made.

Example: 17 counters circled in groups of 4. How many full groups?



Step 1. The model asks how many complete groups fit, so I count groups of the given size and stop when the counters run out rather than dividing straight away.

Step 2. Circling fours uses 4, 8, 12, 16 counters — four full groups — and one counter is left, which is less than 4, so no fifth group is possible.

full groups = 4, remainder 1

Try it: 29 counters in groups of 7. How many are left over?

check: 1

Skill 2 — Deciding What the Remainder Means

The division is the same; the question decides what to do with the remainder.

Drop it — “how many full boxes,” “how many complete rows.”

Add one more group — “how many vans are needed,” “how many boxes to hold them all.”

The remainder IS the answer — “how many are left over.”

Read the question again before you write the number down.

Example: 26 people, tables seat 4. How many tables are needed?

Step 1. Dividing gives a quotient and a leftover, so before writing anything I ask what the leftover people do — and they still need somewhere to sit.

Step 2. $26 \div 4$ is 6 with 2 left over. Six tables seat 24, so the last two people need a seventh table.

tables needed = 7

Try it: Same numbers, different question: 26 people at tables of 4. How many tables are *completely full*?

check: 9

Skill 3 — Estimating Before You Divide

Round the total to a number that divides easily, keep the divisor exact, then divide.

The estimate says roughly how big the quotient should be, so a slip in the long division shows up at once.

Rounding *up* makes the estimate a little too big; rounding *down* makes it a little too small. Knowing which way you went tells you which side the exact answer should land on.

Example: Round 418 to the nearest hundred, ready to estimate $418 \div 4$.

Step 1. I round only the total, because changing the divisor would change the size of the answer completely.

Step 2. Rounding to the hundred means looking at the tens digit, 1; it is below 5, so 418 rounds down.

418 to the nearest hundred is **400**, so the estimate is $400 \div 4 = 100$

Try it: Round 2,740 to the nearest hundred, ready to estimate a division.

0027 : **check: 2700**

(!) Watch out: a remainder is never written as part of the quotient. $23 \div 5$ is “4 remainder 3,” not 43 and not 7. And a remainder equal to or larger than the divisor always means a group was missed.